10.6 Surface Area and Volume Under Dilation

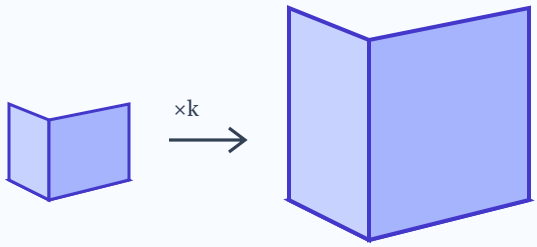
When a solid is dilated (scaled) by factor k, every *linear* measurement (edges, radii, heights, slant heights) scales by $k^1 = k$. Surface area is built from products of two lengths (length × width), so it scales by k^2; volume is built from products of three lengths, so it scales by k^3. This is why doubling a solid's size ($k = 2$) quadruples its surface area but multiplies its volume by 8 — a fact that shows up everywhere from packaging costs to biology (why small animals lose heat faster than large ones, since SA scales slower than volume). This worksheet computes new SA/volume after a dilation, recovers k from a given ratio, and converts between SA ratios and volume ratios.

Examples

Example 1: Scale surface area — and see why k^2

A cube has surface area 54. It is dilated by scale factor $k = 2$. Find the new surface area.

Solution



Dilating a solid by k scales every length by k, so area (length × length) scales by k^2.

Every edge doubles, and SA is built from edge × edge terms, so SA scales by $k^2 = 2^2 = 4$.
New SA $= 54 \times 4 = 216$.
Check with formula directly: original edge s satisfies $6s^2 = 54 \Rightarrow s^2 = 9 \Rightarrow s = 3$. New edge $= 6$, new SA $= 6(6^2) = 216$. ✓
Answer: 216.

Example 2: Scale volume — and see why k^3

A sphere has volume 36π. It is dilated by scale factor $k = 3$. Find the new volume.

Solution

Volume is built from three length factors ($r \times r \times r$), so it scales by $k^3 = 3^3 = 27$.

New $V = 36\pi \times 27 = 972\pi$.

Answer: 972π.

Example 3: Find k from an SA ratio

Two similar solids have surface areas in ratio $9 : 1$ (large to small). Find the scale factor k (large to small).

Solution

SA ratio $= k^2$, so $k^2 = 9$.

$k = \sqrt{9} = 3$ (take the positive root — scale factors are positive).

Answer: $k = 3$.

Example 4: Find k from a volume ratio

Two similar solids have volumes in ratio $8 : 1$ (large to small). Find k.

Solution

Volume ratio $= k^3$, so $k^3 = 8$.

$k = \sqrt[3]{8} = 2$.

Answer: $k = 2$.

Example 5: Scale both SA and volume together

A prism has SA $= 40$ and volume $= 30$. It is dilated by $k = \frac{1}{2}$ (shrunk). Find the new SA and new volume.

Solution

Identify the scale factor: $k = \frac{1}{2}$. Surface area scales by k^2; volume scales by k^3.

Apply each scaling rule to the original measurements:

New SA $= 40 \times k^2 = 40 \times \frac{1}{4} = 10$.

New $V = 30 \times k^3 = 30 \times \frac{1}{8} = \frac{30}{8} = \frac{15}{4}$.

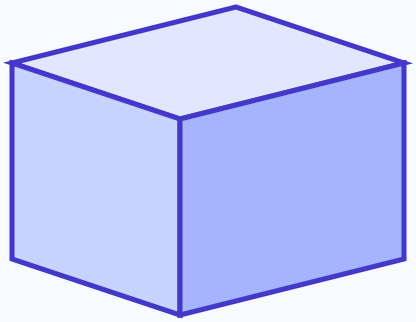
Both measurements shrink, but volume shrinks faster than surface area since it scales by a higher power of k.

Answer: new SA $= 10$; new $V = \frac{15}{4}$.

Example 6: Edge length and SA together

A cube has edge 4 and SA $= 96$. After a dilation with $k = \frac{3}{2}$, find the new edge and new SA.

Solution



Edge scales by k; SA scales by k^2.

New edge $= 4 \times \frac{3}{2} = 6.$

New SA $= 96 \times (\frac{3}{2})^2 = 96 \times \frac{9}{4} = 216.$

Check: $6(6^2) = 216.$ ✓

Answer: new edge $= 6$; new SA $= 216$.

Example 7: Volume ratio from a given k

Two similar cones have scale factor $k = 4$ (large to small). Find the volume ratio (new : old).

Solution

Volume ratio $= k^3 = 4^3 = 64.$

Answer: $64 : 1.$

Example 8: SA ratio from a volume ratio

Two similar solids have volume ratio $27:8$. Find the SA ratio.

Solution

First find k from the volume ratio: $k^3 = \frac{27}{8} \Rightarrow k = \sqrt[3]{\frac{27}{8}} = \frac{3}{2}$.

Then SA ratio $= k^2 = (\frac{3}{2})^2 = \frac{9}{4}$.

Answer: SA ratio $= \frac{9}{4}$, i.e. $9:4$.

Example 9: Shrinking a cylinder

A cylinder has volume 200π. It is dilated by $k = \frac{1}{2}$. Find the new volume.

Solution

Identify the scale factor: $k = \frac{1}{2}$. Volume scales by k^3.

Apply the scaling rule:

New $V = 200\pi \times (\frac{1}{2})^3 = 200\pi \times \frac{1}{8} = 25\pi$.

Shrinking every length by half shrinks the volume to just $\frac{1}{8}$ of the original.

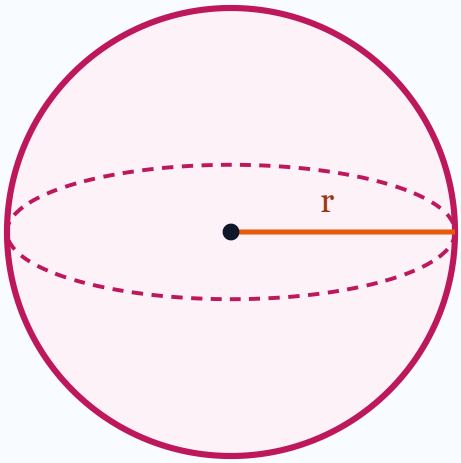
Answer: 25π.

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Example 10: Chain reasoning — volume ratio to k to new SA

A sphere has surface area 144π and volume 288π. A similar (larger) sphere has volume 2304π. Find the scale factor k, then the larger sphere's surface area.

Solution



First recover k from the volume ratio, then apply k^2 to the SA.

Step 1 — find k from the volume ratio: $k^3 = \dfrac{2304\pi}{288\pi} = 8 \;\Rightarrow\; k = \sqrt[3]{8} = 2.$

Step 2 — SA ratio: $k^2 = 4.$

Step 3 — new SA: $144\pi \times 4 = 576\pi.$

Answer: $k = 2$; new SA $= 576\pi$.

Example 11: Real-world application — similar water tanks

A small water tank holds 8 L and needs 2 m^2 of material to build. A larger, similar tank has a linear scale factor of $k = 3$ compared to the small one. Find the large tank's capacity and the material needed to build it.

Solution

Capacity (volume) scales by k^3: $8 \times 3^3 = 8 \times 27 = 216$ L.

Material (surface area) scales by k^2: $2 \times 3^2 = 2 \times 9 = 18$ m^2.

Answer: capacity $= 216$ L; material needed $= 18$ m^2. *Notice the tank holds $27\times$ as much water but needs only $9\times$ the material — bigger containers are more material-efficient per liter.*

Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Scale SA

SA $= 50$, $k = 3$. Find the new SA.

Answer: **Verification:**

Problem 2: Scale volume

Volume $= 12$, $k = 2$. Find the new volume.

Answer: **Verification:**

Problem 3: Find k

SA ratio $16 : 1$. Find k.

Answer: **Verification:**

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Problem 4: Find k

Volume ratio $125:1$. Find k.

Answer: **Verification:**

Problem 5: Shrink SA

$\text{SA} = 180$, $k = \frac{1}{3}$. Find the new SA.

Answer: **Verification:**

Problem 6: Shrink volume

$\text{Volume} = 64\pi$, $k = \frac{1}{2}$. Find the new volume.

Answer: **Verification:**

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Problem 7: SA ratio from k

$k = 5$. Find the SA ratio (new:old).

Answer: **Verification:**

Problem 8: Volume ratio from k

$k = \frac{2}{3}$. Find the volume ratio (new:old).

Answer: **Verification:**

Problem 9: SA ratio from volume ratio

Volume ratio $8 : 27$. Find the SA ratio.

Answer: **Verification:**

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Problem 10: Edge scale

Edge 10, $k = 0.4$. Find the new edge.

Answer: **Verification:**

Problem 11: Both together

SA $= 72$, $V = 48$, $k = 2$. Find the new SA and new V.

Answer: **Verification:**

Problem 12: Find k (reverse direction)

SA ratio $4 : 9$ (small:large). Find k (small to large).

Answer: **Verification:**

Problem 13: Rule

Lengths scale by k. State how SA and volume scale.

Answer: **Verification:**

Problem 14: Word problem

A small object has SA $= 15$ and volume $= 10$. A similar large object has volume $= 80$. Find k and the large object's SA.

Answer: **Verification:**

Problem 15: Challenge

Two similar cones have volumes in ratio $\frac{27}{8}$ (large:small). The small cone has slant height 10 and total SA 75π. Find (a) k, (b) the large cone's slant height, (c) the large cone's SA.

Answer: **Verification:**

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✓ Answer Key

1. 450
2. 96
3. 4
4. 5
5. 20
6. 8π
7. 25 : 1
8. 8 : 27
9. 4 : 9
10. 4
11. SA $= 288$; $V = 384$
12. $\frac{3}{2}$
13. SA by k^2; volume by k^3
14. $k = 2$; SA $= 60$
15. $k = \frac{3}{2}$; slant height 15; SA $= \frac{675\pi}{4}$

Great work — keep practising!